

CO1-ASSIGNMENT I - BASED ON SDG GOALS

Design and Implementation of a Machine Learning Model for Diabetes Prediction

Related SDG: SDG 3 – Good Health and Well-being

Problem Understanding

Diabetes is a chronic disease that affects millions of people worldwide. Early prediction helps doctors diagnose patients quickly and begin treatment at the right time. The objective of this project is to develop a machine learning model that predicts whether a person is diabetic based on medical information.

Dataset

The project uses the Pima Indians Diabetes Dataset. It contains patient information such as pregnancies, glucose level, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, age, and the diabetes outcome.

Use the Pima Indians Diabetes Dataset.

Dataset Features:

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Blood glucose level
BloodPressure	Blood pressure
SkinThickness	Skin fold thickness
Insulin	Insulin level
BMI	Body Mass Index

DiabetesPedigreeFunction	Genetic influence
Age	Age of patient
Outcome	0 = No Diabetes, 1 = Diabetes

Data Collection and Preparation

The dataset is collected, checked for missing or invalid values, cleaned, and prepared for analysis. Proper preprocessing improves the quality of predictions.

Import Libraries

```
import pandas as pd
```

```
import numpy as np
```

Load Dataset

```
df = pd.read_csv("diabetes.csv")
```

Display Data

```
print(df.head())
```

```
print(df.info())
```

```
print(df.describe())
```

Check Missing Values

```
print(df.isnull().sum())
```

Replace Invalid Zero Values

```
cols=['Glucose','BloodPressure','SkinThickness','Insulin','BMI']
```

```
for col in cols:
```

```
df[col]=df[col].replace(0,np.nan)
```

```
df[col].fillna(df[col].median(),inplace=True)
```

Analysis using NumPy and Pandas

NumPy and Pandas are used to summarize the dataset, calculate statistical measures, identify relationships between attributes, and understand the distribution of diabetic and non-diabetic patients.

Shape

```
print(df.shape)
```

Correlation

```
print(df.corr())
```

Diabetes Count

```
print(df['Outcome'].value_counts())
```

Average Values

```
print(df.groupby('Outcome').mean())
```

Visualization and Interpretation

Charts such as histograms, bar charts, and correlation heatmaps help visualize the data and identify important patterns that influence diabetes prediction.

Correlation Heatmap

```
import seaborn as sns
```

```
plt.figure(figsize=(10,8))
```

```
sns.heatmap(df.corr(),annot=True,cmap="coolwarm")
```

```
plt.show()
```

Concept Learning

The model learns general rules from training examples to classify new patient records as diabetic or non-diabetic.

Split Dataset

Python

Run

```
from sklearn.model_selection import train_test_split
```

```
X=df.drop("Outcome",axis=1)
```

```
y=df["Outcome"]
```

```
X_train,X_test,y_train,y_test=train_test_split(
```

```
X,y,test_size=0.2,random_state=42)
```

Version Spaces and Candidate Elimination

Version space represents all hypotheses consistent with the training data. Candidate Elimination gradually removes inconsistent hypotheses until only valid ones remain.

Inductive Bias

The learning algorithm assumes that the patterns learned from the training data will also apply to unseen patient records.

Decision Tree Learning

A Decision Tree classifies patients by selecting the most informative attributes and creating decision rules that lead to the final prediction.

Representation Algorithm

The learned knowledge is represented as a tree structure consisting of decision nodes and leaf nodes, making the model easy to understand.

Glucose > 125?

Yes

/ \

BMI>30 No

/ \ \

Yes No Non-Diabetic

Heuristic Space Search

Decision Tree learning uses Information Gain as a heuristic to select the best attribute at each stage of tree construction.

Example Output

Accuracy : 76%

Precision : 75%

Recall : 74%

SDG Mapping

This project supports SDG 3 (Good Health and Well-being) by enabling early disease detection, improving healthcare decisions, reducing complications, and promoting preventive healthcare.

This project supports SDG 3 by:

- Early detection of diabetes
- Reducing health risks
- Assisting doctors in diagnosis
- Improving patient care
- Promoting preventive healthcare

Conclusion

The diabetes prediction model demonstrates how machine learning can assist healthcare professionals in making faster and more accurate decisions. Proper data preparation, analysis, and interpretation improve the reliability of predictions and contribute to better public health.